

DASE7503 Robotic Systems Integration

Group Project – A Smart Wheel-driven Robot for Logistics Applications

(40% of Total Course Mark)

Objectives

This group project is aimed at providing a hands-on experience for students to apply their knowledge and skills, such as robotic systems design, control, ROS programming, etc., to design and develop a smart wheel-driven robot for logistics applications with integration of ESP32 & Raspberry Pi boards, lidar, camera, etc.

A group of around 8 students designs and builds a smart wheel-driven robot with a lifting device for locating a cargo rack and then delivering it to a specific zone.

Each group is required to submit the following:

- SolidWorks CAD assembly models of a smart wheel-driven robot
- All computer programs
- A physical robot assembly

Major equipment, software and materials

- Equipment
 - 3D printers
 - Laser material processing equipment
- Software
 - SolidWorks
 - Arduino IDE
 - ROS 2 Jazzy
 - Python
- Materials
 - Acrylic board
 - 3D printing materials
 - ESP32 and Raspberry Pi development boards
 - Lidar, motors, cameras

DASE7503 机器人系统集成

小组项目——面向物流应用的智能轮式机器人

(占课程总成绩的 40%)

目标

本小组项目旨在为学生提供实践机会，运用所学的知识与技能，如机器人系统设计、控制、ROS 编程等，设计并开发一台面向物流应用的智能轮式机器人（smart wheel-driven robot），并集成 ESP32 与树莓派（Raspberry Pi）开发板、激光雷达（lidar）、摄像头等。

每组分 8 名学生，设计并制作一台带有升降装置（lifting device）的智能轮式机器人，用于定位货架（cargo rack）并将其运送至指定区域。

每组须提交以下内容：

- 智能轮式机器人的 SolidWorks CAD 装配体模型
- 全部计算机程序
- 一台实体机器人装配体

主要设备、软件 and 材料

- 设备
 - 3D 打印机
 - 激光材料加工设备
- 软件
 - SolidWorks
 - Arduino IDE
 - ROS 2 Jazzy
 - Python
- 材料
 - 亚克力板
 - 3D 打印材料
 - ESP32 与树莓派开发板
 - 激光雷达、电机、摄像头

Procedures

1. Design and create a 3D assembly model of a smart wheel-driven robot using SolidWorks CAD software.
2. Adopt 3D printers and laser material processing equipment to fabricate the required mechanical/housing parts of the robot
3. Assemble the required parts, a lidar, and a camera to form a complete robot.
4. Write the required control programs (micro-ROS) on ESP32 to control the movement of the robot and the movement of the lifting device on the robot.
5. Write the required programs on Raspberry Pi (ROS2 Jazzy) to do the following:
 - a. control a robot's camera to scan a QR code and store the decoded QR code data.
 - b. control a robot's lidar for localisation and navigation
 - c. communicate with the ESP32 via a USB serial port
6. Write a program on either a Raspberry Pi or a PC to achieve a robot's autonomous navigation.
7. Test the robot.

Tasks for each group

Each group should complete the following:

1. Design and fabricate a smart wheel-driven robot with integration of a lifting device.
2. Develop all required programs for achieving the robot's functionalities, including lifting, QR scanning, localisation and navigation, and the robot's autonomous navigation, etc.
3. Participate in a robot (game) demonstration. Within 3 minutes, the robot needs to navigate in the game field to locate and collect the cargo rack at different points of the game field and transfer each cargo rack to the destination point.

Project requirements

Each group should pay attention to the following:

- The overall size of a smart robot (physical robot) should not be larger than 30cm x 30cm x 18cm.
- Only one set of required components will be provided for each group.
- Return all group project components, tooling, and materials (not assembled in the physical robot) on Nov 26, 2026.

Details of the game

i) Game field

- The field size is around 300cm x 200cm.
- All points on the field, including the start point, destination point and cargos, have their own QR codes for route marking, as shown in Figure 1.
- Cargo racks' locations are fixed on the field. All participants also know their locations (x, y) in advance.

ii) Game rules

流程

1. 使用 SolidWorks CAD 软件设计并创建智能轮式机器人的三维装配体模型。
2. 采用 3D 打印机和激光材料加工设备，制作机器人所需的机械/外壳零件。
3. 将所需零件、激光雷达和摄像头组装成一台完整的机器人。
4. 在 ESP32 上编写所需的控制程序（micro-ROS），以控制机器人的运动及其升降装置的运动。
5. 在树莓派（ROS2 Jazzy）上编写所需程序，实现以下功能：
 - a. 控制机器人的摄像头扫描二维码（QR code）并存储解码后的二维码数据。
 - b. 控制机器人的激光雷达进行定位与导航
 - c. 通过 USB 串口与 ESP32 通信
6. 在树莓派或 PC 上编写程序，实现机器人的自主导航。
7. 对机器人进行测试。

各组任务

每组须完成以下工作：

1. 设计并制作一台集成升降装置的智能轮式机器人。
2. 开发实现机器人各项功能所需的全部程序，包括升降、二维码扫描、定位与导航，以及机器人的自主导航等。
3. 参加机器人（比赛）演示。机器人须在 3 分钟内于比赛场地中导航，定位并收集位于场地不同点位的货架，并将每个货架运送至目的点。

项目要求

各组须注意以下事项：

- 智能机器人（实体机器人）的整体尺寸不得大于 30cm x 30cm x 18cm。
- 每组仅提供一套所需元器件。
- 所有小组项目元器件、工具及材料（未装配在实体机器人上的部分）须于 2026 年 11 月 26 日归还。
译者注：此处归还日期为 2026 年 11 月 26 日，而第 4 页“提交”部分要求工具及材料于 2026 年 11 月 30 日 17:00 前提交，两处日期不一致，请以课程方确认为准。

比赛详情

i) 比赛场地

- 场地尺寸约为 300cm x 200cm。
- 场地上的所有点位，包括起点、目的点和货物，均设有各自的二维码作为路线标记，如图 1 所示。
- 货架在场地上的位置固定。所有参赛者也会预先知道其位置坐标 (x, y)。

ii) 比赛规则

1. The game is based on groups. Each group is considered as one participant.
2. Each group's robot should use its camera to scan a QR code of the start point, and the robot should start identifying and picking a cargo rack in an automatic way. At the same time, the game's timer starts counting.
3. Using a lidar, the robot should automatically navigate to a zone with a cargo rack and use its camera to scan a QR label assigned to the rack to locate the rack and lifting and delivering it to the destination area.
4. Within 3 minutes, Step 3 is repeated until all racks are delivered to the destination area.
5. The rack cannot touch any floor during delivery.
6. If either a robot or a rack collides with a wall, 2 points will be deducted for each collision.
7. At the end of 3 minutes' game time, the number of cargo racks that lay in the destination area will be recorded. 5 points will be awarded for each shipment, and the final score will be tied to the ranking of all groups.

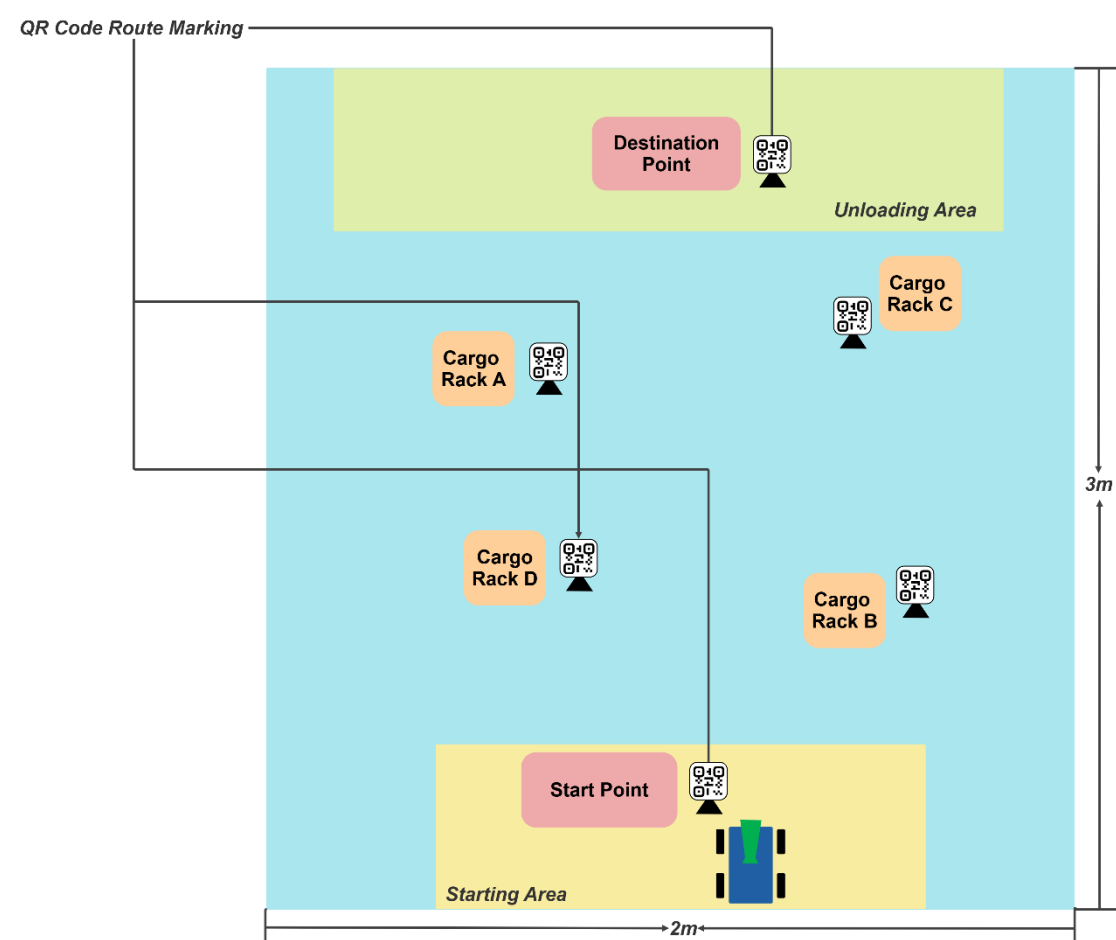


Figure 1. Game field

Assessment

The group project assesses the overall performance of each student based on his/her participation in and contributions to the project. The assessment consists of two main parts:

- | | |
|------------------------------------|-----------------------------------|
| I. Overall Group Assessment | (25% of Total Course Mark) |
| II. Game Demonstration | (10% of Total Course Mark) |

1. 比赛以小组为单位进行，每个小组视为一个参赛者。
2. 各组机器人须使用摄像头扫描起点的二维码，随后以自动方式开始识别并拾取货架，比赛计时器同时开始计时。
3. 机器人须借助激光雷达自动导航至有货架的区域，并使用摄像头扫描该货架上的二维码标签以定位货架，然后将其抬起并运送至目的区域。
4. 在 3 分钟内重复步骤 3，直至所有货架都运送至目的区域。
5. 运送过程中货架不得接触地面。
6. 机器人或货架每碰撞墙壁一次，扣 2 分。
7. 3 分钟比赛时间结束时，记录位于目的区域内的货架数量。每成功运送一个货架得 5 分，最终得分将与各组排名挂钩。

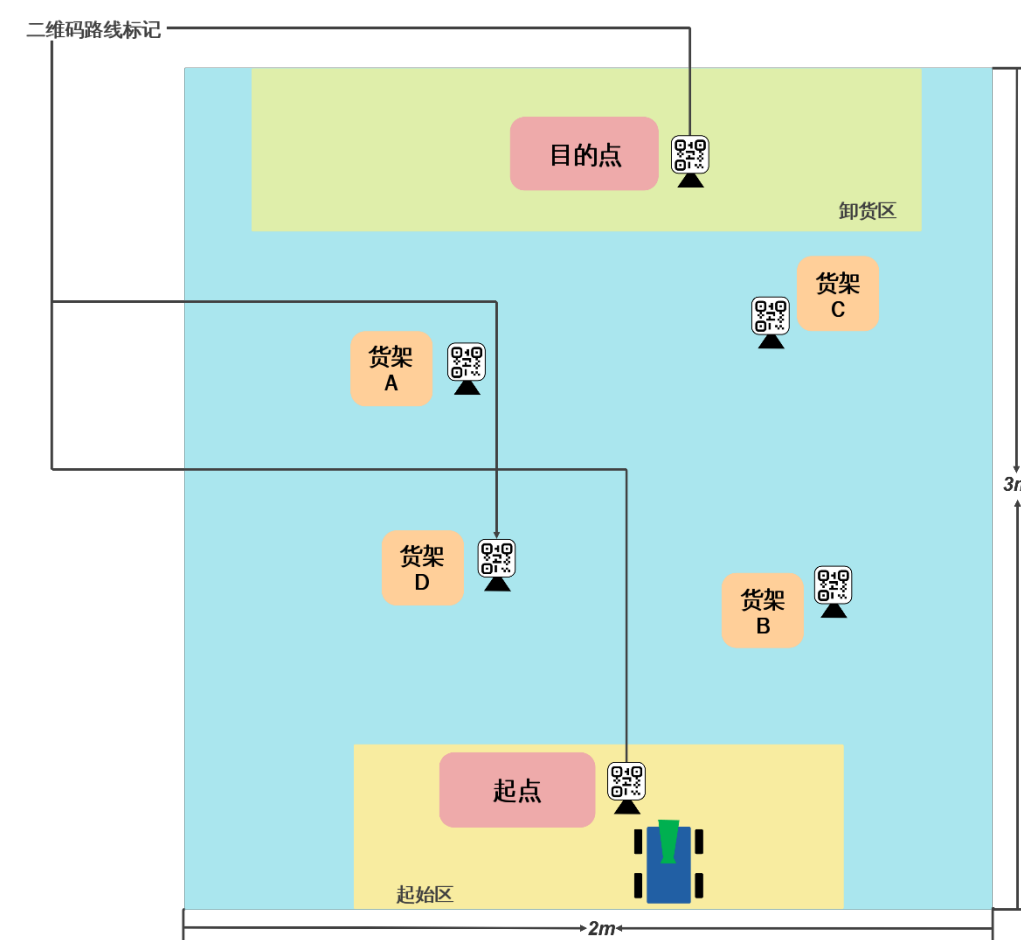


图 1. 比赛场地

评估

本小组项目根据每位学生对项目的参与程度和贡献来评估其整体表现。评估包括两个主要部分〔译者注：下文实际列出 I、II、III 共三个部分〕：

- | | |
|------------------|----------------------|
| I. 小组整体评估 | (占课程总成绩的 25%) |
| II. 比赛演示 | (占课程总成绩的 10%) |

III. Peer Assessment (5% of Total Course Mark)

I. Overall Group Assessment (25% of Total Course Mark)

The teacher will be responsible for this overall group assessment. The performance of each group is based on the following criteria:

- (a) **Design concept and innovation**
 - Creativity
 - Originality
- (b) **Robot product details**
 - Aesthetics (visual appearance)
 - Functionality
 - Easiness of assembly
- (c) **Programs**
 - Performance
 - Easiness of code reading
 - User-friendliness

II. Game Demonstration (10% of Total Course Mark)

Each group is assessed by the performance of each robot's demonstration with the following criteria:

- Achieve automatic navigation and complete all the operations within a specific time (A range)
- Achieve manual control and complete all the operations within a specific time (B range)

III. Peer Assessment (5% of Total Course Mark)

Each group member is assessed by other group members to award each group member's contributions to the group project.

Submission

Each group is required to submit the following items:

1. A completed and tested Smart Wheel-driven Robot;
2. All CAD models and files, including a SolidWorks design zip file created by the "pack-and-go" function;
3. All control programs, including ROS2 Jazzy files and micro-ROS files; and
4. Group Project tooling and materials, including plastic box, display cables, Raspberry Pi 5 27W power supply, USB 3.2 card readers, battery charger, power adaptor, unused parts;

Items 1 and 4 should be submitted to **Mr. Derek Tong** or **Mr. Mark Wan** in the Laboratory **HW 103A** by **17:00 on Nov 30, 2026 (Monday)**.

Besides, items 2-3 should be submitted to the Moodle by **23:59 on Nov 30, 2026**.

III. 同伴互评 (占课程总成绩的 5%)

I. 小组整体评估 (占课程总成绩的 25%)

小组整体评估由任课教师负责。各组表现依据以下标准评定：

- (a) **设计理念与创新**
 - 创造性
 - 原创性
- (b) **机器人产品细节**
 - 美观度 (外观)
 - 功能性
 - 易装配性
- (c) **程序**
 - 性能
 - 代码易读性
 - 用户友好性

II. 比赛演示 (占课程总成绩的 10%)

各组根据其机器人演示的表现进行评估，标准如下：

- 实现自动导航，并在规定时间内完成全部操作 (A 档)
- 实现手动控制，并在规定时间内完成全部操作 (B 档)

III. 同伴互评 (占课程总成绩的 5%)

每位组员由其他组员评估，以评定每位组员对小组项目的贡献。

提交

每组须提交以下各项：

1. 一台已完成并经过测试的智能轮式机器人；
2. 全部 CAD 模型及文件，包括使用“pack-and-go” (打包) 功能生成的 SolidWorks 设计 zip 文件；
3. 全部控制程序，包括 ROS2 Jazzy 文件和 micro-ROS 文件；以及
4. 小组项目工具及材料，包括塑料盒、显示器线缆、Raspberry Pi 5 27W 电源、USB 3.2 读卡器、电池充电器、电源适配器以及未使用的零件；

第 1 项和第 4 项须于 **2026 年 11 月 30 日 (星期一) 17:00** 前提交至 **HW 103A** 实验室的 **Mr. Derek Tong** 或 **Mr. Mark Wan**。

此外，第 2-3 项须于 **2026 年 11 月 30 日 23:59** 前提交至 Moodle。